

Research Article

Impact of Rice Husk Ash Properties on Concrete Strength: Experimental and Machine Learning Study

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Received 24 September 2024; Accepted 10 March 2025

Academic Editor: Mustafa Zeybek

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This study explores the substitution of cement with rice husk ash in concrete, aiming to reduce the pollution and energy consumption caused by cement production. Through a series of experiments, including slump test, water absorption test, compressive strength test, and dynamic compression test, the effects of calcination temperature and average particle size of rice husk ash on concrete performance were evaluated. The findings indicate that the use of rice husk ash lowers slump while increasing the concrete's water absorption capacity. With higher calcination temperatures, both compressive and dynamic compressive strengths initially improve, followed by a decline. Smaller rice husk ash particles lead to better performance in these strength tests. The ideal parameters identified were a calcination temperature of 650°C and a particle size of 5 µm. A machine learning approach, utilizing the eXtreme Gradient Boosting (XGB) model, was employed to predict dynamic compression strength, and the predictions were cross-referenced with experimental data. Shapley analysis was applied to assess the influence of individual variables on the strength outcomes. Results confirmed the accuracy of the XGB model, with an R^2 value of 0.9435 and a mean absolute error (MAE) of 2.29 MPa, with particle size showing a more significant impact on dynamic compressive strength than calcination temperature.

Keywords: compressive strength; concrete; dynamic compressive strength; machine learning; particle sizes; rice husk ash; slump; sustainable construction; varying calcination temperatures; water absorption rate

1. Introduction

In the construction industry, concrete is one of the main structural materials and the cornerstone of modern infrastructure construction [1]. The preparation of concrete requires a large amount of cement, which is usually calcined at high temperature, which not only consumes a lot of energy but also releases a large amount of greenhouse gases such as carbon dioxide, causing serious negative impact on the environment [2]. By 2023, the global cement production will exceed 4 billion tons. Studies have shown that a ton of cement production will emit 0.8 tons of carbon dioxide gas [3]. Therefore, reducing the production and use of cement is of great significance to environmental protection.

Over the years, numerous efforts have been made by researchers to mitigate the environmental pollution caused by cement production, including the development of low-carbon

alternatives and the incorporation of industrial by-products as raw materials. The integration of supplementary cementitious materials (SCMs) to partially replace cement has also been extensively studied [4–6]. It should be noted that SCM are usually industrial by-products, so the use of SCMs to replace part of the cement can help reduce the exploitation and consumption of natural resources and achieve resource recycling [7]. Additionally, the relatively low cost of these materials can contribute to reducing the production costs of concrete [8]. Traditional SCM include fly ash, slag, and silica fume. Li et al. [9] reviewed the advantages and disadvantages of fly ash as a supplementary material, highlighting its ability to reduce concrete porosity and enhance impermeability. However, the high alkalinity of fly ash may lead to alkali-silica reactions, negatively affecting the long-term durability of concrete. Lothenbach et al. [10] reported that while slag has a

TABLE 1: Mix ratio design and sample label.

Specimen	Cement (kg/m ³)	Coarse aggregate (kg/m ³)	Sand (kg/m ³)	Water (kg/m ³)	Calcination temperature of rice husk ash (°C)	Average particle size of rice husk ash (μm)
C	375	1202	625	210	—	—
C450-5	337.5	1202	625	210	450	5
C550-5	337.5	1202	625	210	550	5
C650-5	337.5	1202	625	210	650	5
C750-5	337.5	1202	625	210	750	5
C450-25	337.5	1202	625	210	450	25
C550-25	337.5	1202	625	210	550	25
C650-25	337.5	1202	625	210	650	25
C750-25	337.5	1202	625	210	750	25

lower hydration heat, making it suitable for mass concrete, its performance depends on the type of activator used, necessitating careful design of the concrete mix. Ahmad et al. [11] pointed out that an appropriate amount of silica fume is conducive to improving the strength and durability of concrete, but as it is a by-product of ferrosilicon alloy production, the cost of silica fume is extremely high, which is not conducive to large-scale application.

In recent years, rice husk ash has gained significant attention as a novel SCM [12]. Unlike traditional materials, rice husk ash offers distinct advantages: (1) It contains a high percentage of silica (85%–95%), which imparts strong pozzolanic reactivity [13, 14]; (2) being a by-product of agriculture, rice husk is both renewable and sustainable [15]; and (3) by adjusting the incineration temperature and treatment methods, the physical and chemical properties of rice husk ash can be modified to enhance the performance of concrete with varying needs [16, 17]. Previous research has extensively explored the behavior of concrete incorporating rice husk ash in various aspects. Wu et al. [14] studied the influence of different contents of rice husk ash on the performance of concrete and found that when rice husk ash replaced 60% of cement, the local microstructure was the densest, but the excessive content of rice husk ash caused dilution effect on cement, resulting in a decrease in the strength of concrete. Camargo-Perez et al. [18] observed that increasing rice husk ash content reduced the workability of concrete, as shown by a progressive slump reduction. Madandoust et al. [19] highlighted that while the early strength of concrete with rice husk ash was lower than that of conventional concrete, its long-term strength was significantly superior. However, the influence of rice husk ash on concrete porosity remains inconclusive, as findings vary depending on particle fineness and chemical composition of the ash [20]. Regarding the durability of rice husk ash concrete, Yao et al. [21] demonstrated that under sulfate exposure, rice husk ash concrete exhibited better corrosion resistance compared to ordinary concrete. Additionally, Zhang et al. [22] showed that using rice husk ash at replacement levels of 10%, 20%, and 30% significantly reduced pore connectivity and delayed structural degradation during freeze–thaw cycles.

In summary, although extensive studies have been conducted on the properties of rice husk ash concrete, the effects

of calcination temperature and particle fineness on concrete performance remain unclear. Therefore, this study focuses on rice husk ash produced at varying calcination temperatures and sieved into different particle sizes for partial cement replacement. Slump and water absorption tests were performed, alongside compressive strength and dynamic compression experiments, to analyze the changes in both static and dynamic compressive strengths. Additionally, a dynamic compressive strength prediction model for rice husk ash concrete was developed using machine learning techniques. This study aims to provide insights and references for the practical application of rice husk ash concrete. In addition, the multidimensional analysis combining rice husk ash calcination temperature, particle size, and machine learning models represents an underexplored direction in the field.

2. Materials and Methodology

2.1. Constituent Materials. Ordinary Portland cement (P·O 42.5) was utilized, along with natural gravel as the coarse aggregate and river sand as the fine aggregate. Rice husk ash was sourced from a processing facility in Tianjin, China, and subjected to calcination at 450°C, 550°C, 650°C, and 750°C, resulting in particle sizes of 5 and 25 μm, to investigate the impact of varying levels of silica reactivity and carbon content removal on RHA properties and to explore the influence of fine and coarse particles on concrete properties.

2.2. Proportioning and Sample Preparation. Under normal circumstances, when the replacement ratio of rice husk ash is 10%, the strength and working performance of concrete can be well balanced. In this research, rice husk ash replaced 10% of the cement by weight. Nine concrete sample groups were prepared, with the specific mix ratios and sample labels outlined in Table 1.

2.3. Experimental Procedures. In this research, samples were placed in a curing chamber maintained at a constant temperature of $20 \pm 2^\circ\text{C}$ and a relative humidity of 95% for the entire curing duration. These conditions align with ASTM C511 standards for concrete curing. Nine groups of concrete samples were tested for water absorption and compressive strength over curing periods of 3, 7, 28, 56, and 90 days. The dynamic compression test was conducted at 28 days of

curing. Additionally, the slump of the freshly mixed concrete was evaluated.

Slump testing was performed during the mixing process to assess the impact of various calcination temperatures and rice husk ash particle sizes on the workability of concrete. The water absorption of the nine sample groups was measured at different curing intervals, following the formula shown in Equation (1):

$$\omega = \frac{m_0 - m_1}{m_1} \times 100\%, \quad (1)$$

where ω is the water absorption (%), m_0 is the mass of the specimen after immersion (kg), and m_1 is the mass of the specimen after drying (kg).

For the compressive strength test, a WAW-2000D electro-hydraulic servo universal testing machine was used. The sample dimensions were 150 mm × 150 mm × 150 mm, and the testing speed was set to 3 mm/min.

The dynamic compression test employed a Hopkinson rod system, each with a diameter of 36 mm and a length of 1.5 m, with samples shaped as cylinders (36 mm in height and 74 mm in diameter). The impact pressures (0.25 to 0.55 MPa) were selected to simulate the dynamic compressive behavior of concrete under high strain rate conditions that may occur during seismic events, explosions, or high-speed impacts. These pressures fall within the range commonly used in Hopkinson bar tests to evaluate the dynamic performance of construction materials. The impact pressures were set at 0.25, 0.35, 0.45, and 0.55 MPa. Before each dynamic compression test, Vaseline was applied to the sample surface to ensure smooth contact between the specimen and the Hopkinson bar.

3. Results and Discussion

3.1. Slump. The results of the slump test for the freshly mixed concrete are illustrated in Figure 1. It is observed in Figure 1 that replacement of cement by rice husk ash reduces the slump of the newly mixed mixture. This is consistent with the results of previous studies [23–25]. This is mainly due to the fact that the specific surface area of rice husk ash is larger than that of cement, so the total specific surface area of the mixture is larger after replacing cement, reducing the free water in the mixture and resulting in lower slump [23]. Additionally, as shown in Figure 1, the slump gradually decreases with higher calcination temperatures. For instance, with a particle size of 5 μm and a calcination temperature of 550°C, the slump is reduced by 30% compared to ordinary concrete. Similarly, with a particle size of 25 μm and a calcination temperature of 650°C, the slump decreases by 23.37%. This effect is likely due to the creation of more microporous structures within the rice husk ash, which increases specific surface area and reduces fluidity.

At the same calcination temperature, the slump of mixtures with a 25 μm particle size is higher than those with a 5 μm particle size. For calcination temperatures of 450°C, 550°C, 650°C, and 750°C, the slump values of the 25 μm

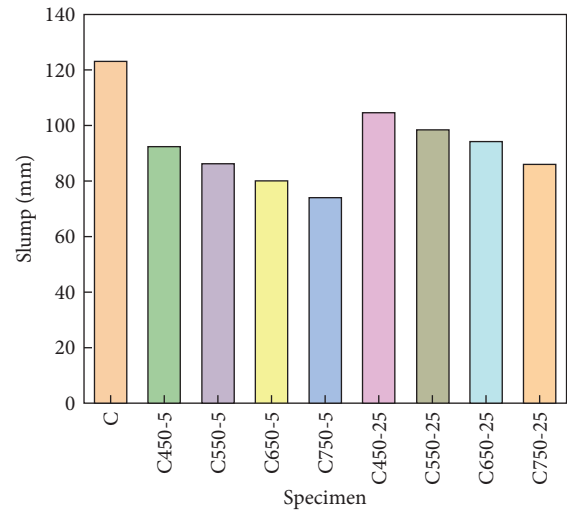


FIGURE 1: Slump value.

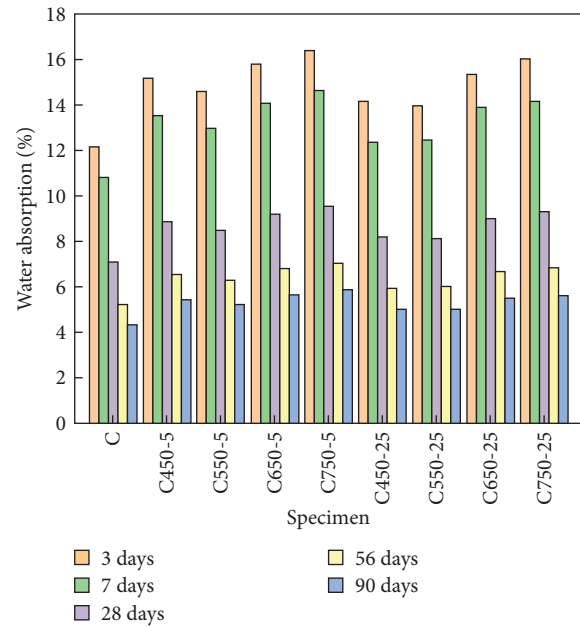


FIGURE 2: Water absorption test results.

particle size increased by 13.33%, 14.29%, 17.89%, and 24.47%, respectively, compared to the 5 μm particle size. These findings suggest that smaller grain sizes of rice husk ash result in lower slump, primarily because smaller particles fill voids between cement grains more effectively, but they also reduce the free water content within the mix [26].

3.2. Water Absorption. The water absorption values of the nine sample groups at various curing ages are displayed in Figure 2. It is evident from Figure 2 that the water absorption of concrete containing rice husk ash is higher than that of ordinary concrete. This observation aligns with previous studies [27, 28]. According to Ijabo and Itodo [29], rice husk ash is essentially a hygroscopic material, so it will increase the water absorption of concrete. The multilayer porous structure

of rice husk ash also plays a significant role in raising water absorption [21]. As curing progresses, the water absorption of both ordinary and rice husk ash concrete gradually decreases, mainly due to the continuous cement hydration reaction, which produces hydration products like calcium silicate gel that fill the concrete's internal pores, increasing its hardness and compactness and reducing water absorption [30]. It was also noted that, with the same particle size of rice husk ash, a rise in calcination temperature initially leads to a reduction in water absorption, followed by an increase. This occurs because the activity of rice husk ash is lower at lower calcination temperatures, resulting in greater porosity. However, at optimal calcination temperatures, the ash's activity increases, helping to fill the pores and further reduce the water absorption rate. When the calcination temperature is too high, rice husk ash may form crystals, increasing porosity and water absorption. Moreover, smaller particle sizes of rice husk ash generally show higher water absorption rates at the same calcination temperature, as they provide a larger surface area for water molecules, facilitating greater absorption. However, higher water absorption can lead to increased permeability, which may allow harmful substances such as chlorides and sulfates to penetrate the concrete matrix more easily, thus accelerating deterioration.

3.3. Compressive Strength. The compressive strength values for the nine sample groups at various curing ages are presented in Figure 3. As shown in Figure 3, the compressive strength of the samples increases progressively with curing age. For the C450-5, C450-25, and C750-25 samples, compressive strength remains lower than that of ordinary concrete at curing ages of 3 and 7 days. Even at 28, 56, and 90 days, the compressive strength of C450-25 and C750-25 samples remains below that of the control concrete. Additionally, Figure 3 illustrates that when the particle size of rice husk ash is constant, compressive strength initially increases and then decreases as the calcination temperature rises. The highest compressive strength is observed when the calcination temperature reaches 650°C. Moreover, the growth rate of compressive strength for C650-5 and C650-25 samples, as curing progresses, is significantly higher than that of the ordinary concrete.

It is crucial to mention that an increase in the particle size of rice husk ash has a detrimental effect on compressive strength. The reasons are as follows: (1) At 450°C, rice husk ash contains significant amounts of unburned carbon and other organic materials, which lowers its reactivity, leading to a reduction in compressive strength. (2) When the calcination temperature reaches 550°C, most organic substances are removed, resulting in moderate improvements in specific surface area and pozzolanic activity, but these improvements are insufficient to markedly boost early and long-term strength. (3) At 650°C, the ash predominantly consists of active silica, which maximizes specific surface area, resulting in the formation of more hydration products and an enhancement in both early and long-term strength. (4) At 750°C, the reactivity of the ash declines because the higher temperature

causes the silica to crystallize, decreasing both surface area and pozzolanic potential. (5) The smaller particle size of rice husk ash enables rice husk ash to better fill the internal pores of concrete, provide a larger reaction area, and generate more hydration products, which enhances the structural stability and strength of concrete.

Although the reduced water absorption at 550°C suggests a denser microstructure, the optimal compressive strength is achieved at 650°C. This apparent contradiction can be explained by the higher pozzolanic activity of rice husk ash at 650°C. At this temperature, the amorphous silica content reaches its peak reactivity, leading to the formation of additional C-S-H gel, which contributes significantly to strength development. In contrast, at 550°C, while the microstructure is denser, the lower reactivity of the ash limits the production of hydration products, resulting in relatively lower compressive strength. Thus, the results highlight the need to consider multiple factors when evaluating the performance of rice husk ash concrete.

3.4. Dynamic Compression Performance

3.4.1. Dynamic Compression Strength. The dynamic compressive strength values for the nine sample groups under varying impact pressures are presented in Figure 4. As illustrated in Figure 4, dynamic compressive strength increases progressively with higher impact pressures. Previous studies on various types of concrete have shown similar results [31–34]. This is mainly because as the impact pressure increases, the strain rate brought by the impact load will increase, and a higher strain rate will make the local stress concentration more obvious, and the stress in the stress concentration area is higher, so the dynamic compression strength will gradually increase [35].

Additionally, it was observed that the dynamic compressive strength of C450-5, C450-25, and C750-25 samples is lower than that of ordinary concrete under different impact pressures, whereas other samples demonstrated higher dynamic compressive strength, with C650-5 showing the best performance. For instance, at an impact pressure of 0.65 MPa, the dynamic compressive strength of C450-25 and C750-25 was reduced by 3.51% and 10.35%, respectively, compared to ordinary concrete.

This trend is consistent with the compressive strength results, where the dynamic compressive strength of rice husk ash concrete initially increases, followed by a decrease as calcination temperature rises. Notably, the dynamic compressive strength of C750-5 exceeded that of C450-5, while the dynamic compressive strength of C750-25 was significantly lower than that of C450-25. Several factors contribute to these results: (1) Although rice husk ash's reactivity diminishes at higher calcination temperatures, the smaller particle sizes result in a higher specific surface area, allowing for more effective participation in cement hydration. Therefore, even with reduced reactivity, the filling effect of rice husk ash remains strong, improving dynamic compressive strength. (2) The increase of the average particle size of rice husk ash

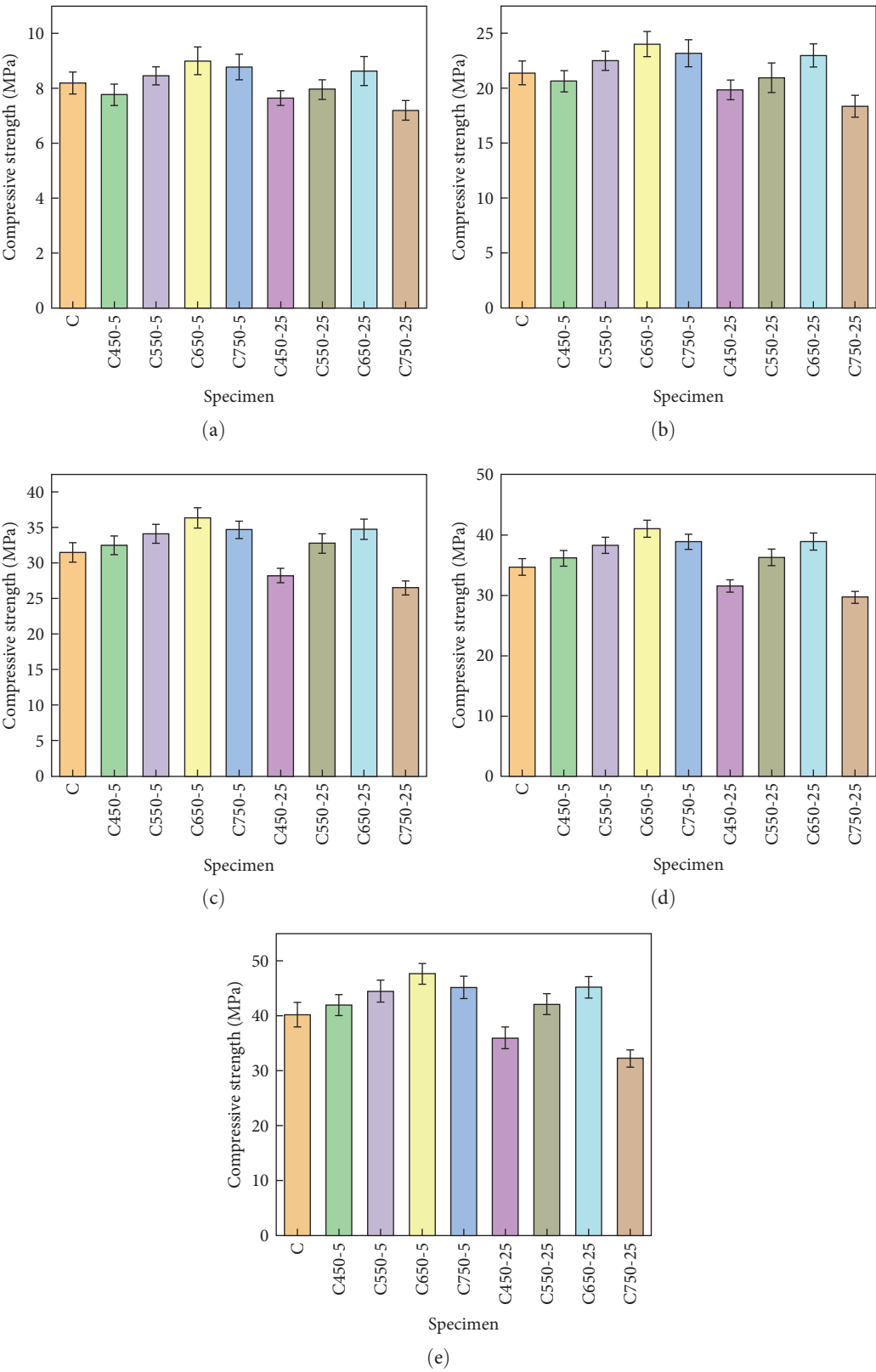


FIGURE 3: Compressive strength values at different curing ages: (a) 3 days, (b) 7 days, (c) 28 days, (d) 56 days, and (e) 90 days.

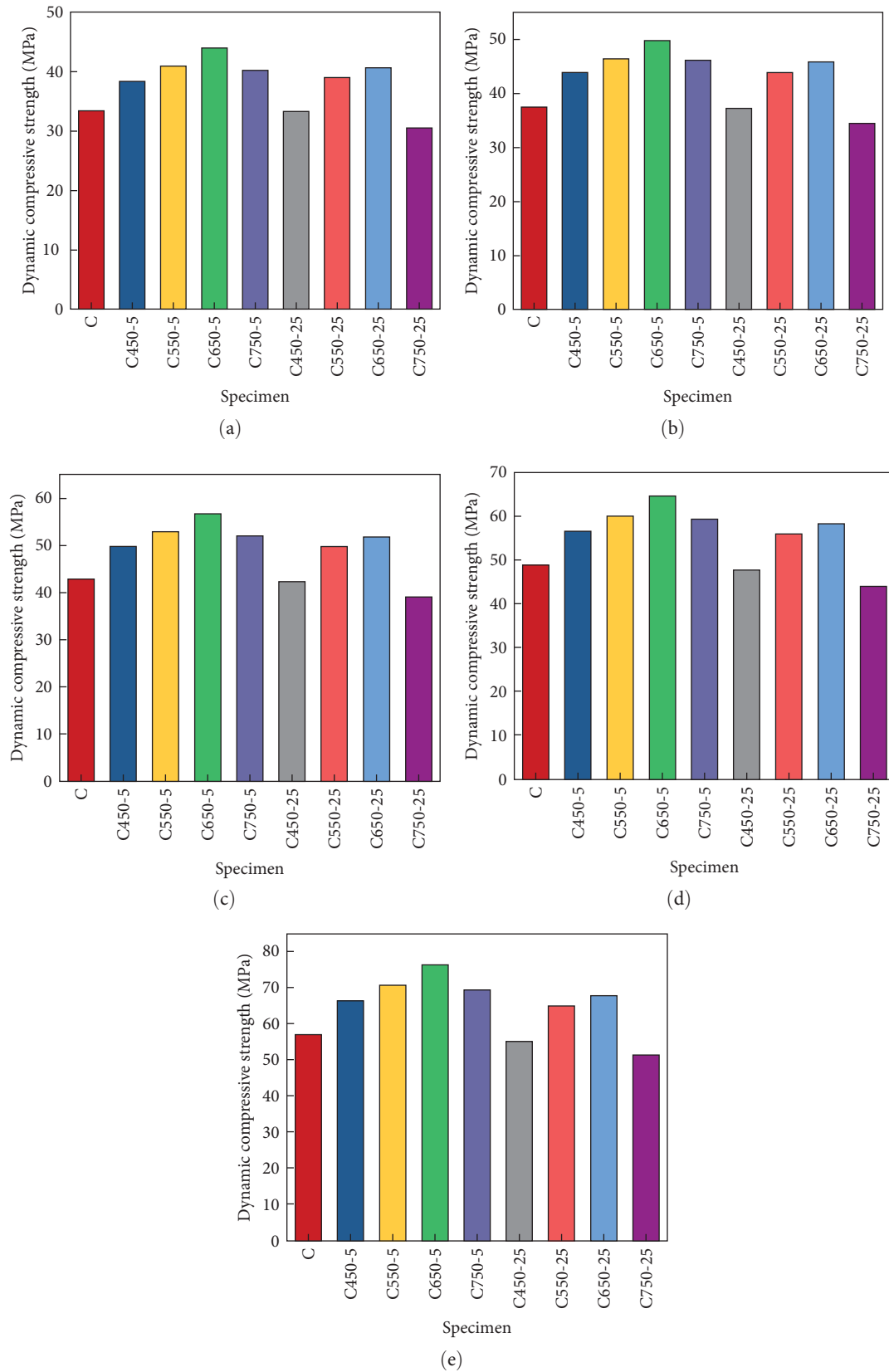


FIGURE 4: Dynamic compressive strength values at different impact pressures: (a) 0.25 MPa, (b) 0.35 MPa, (c) 0.45 MPa, (d) 0.55 MPa, and (e) 0.65 MPa.

TABLE 2: Results of DIF value calculation.

Specimen	DIF				
	0.25 MPa	0.35 MPa	0.45 MPa	0.55 MPa	0.65 MPa
C	1.06	1.19	1.36	1.55	1.81
C450-5	1.18	1.35	1.53	1.74	2.04
C550-5	1.2	1.36	1.55	1.76	2.07
C650-5	1.21	1.37	1.56	1.78	2.1
C750-5	1.16	1.33	1.5	1.71	2.0
C450-25	1.18	1.32	1.5	1.69	1.95
C550-25	1.19	1.34	1.52	1.71	1.98
C650-25	1.17	1.32	1.49	1.68	1.95
C750-25	1.15	1.30	1.47	1.66	1.93

reduces its specific surface area and filling effect, resulting in that rice husk ash cannot fully participate in the hydration reaction but increases the porosity, thus making the dynamic compressive strength of C750-25 lower than that of C450-25.

3.4.2. Dynamic Impact Factor. The dynamic increase factor (DIF) is commonly used to describe the strain rate effects on concrete under dynamic impact loads. The DIF values for the nine sample groups subjected to different impact pressures are presented in Table 2. It is evident from Table 2 that the DIF is closely linked to the impact pressure, with the DIF gradually increasing as impact pressure rises. Additionally, the DIF values of the samples containing rice husk ash are consistently higher than those of ordinary concrete. Previous studies have concluded that concrete is sensitive to strain rates [36, 37]. Consequently, rice husk ash enhances the strain rate sensitivity of concrete under dynamic loading conditions. This is mainly because rice husk ash particles can exert reinforcement effect at different scales, especially from nanoscale pore filling to micron-scale particle bridging, which makes concrete exhibit higher strain rate sensitivity under dynamic impact load.

Furthermore, the calcination temperature of rice husk ash and its average particle size significantly affect the DIF. Specifically, as the calcination temperature increases, the DIF initially rises and then declines, while the DIF of rice husk ash decreases as particle size increases.

Liang et al. [38] believed that there was an obvious logarithmic relationship between DIF value and strain rate and gave an empirical formula, as shown in Equation (2). The DIF value obtained by the experiment is fitted according to Formula (2). The comparison between the fitting results and the experimental results is shown in Figure 5, and the expression is shown in Formula (3). It can be seen that there is little difference between the measured DIF value and the fitting result, indicating that Equation (3) can express the relationship between DIF and strain rate:

$$\text{DIF} = a \lg \dot{\epsilon} + b, \quad (2)$$

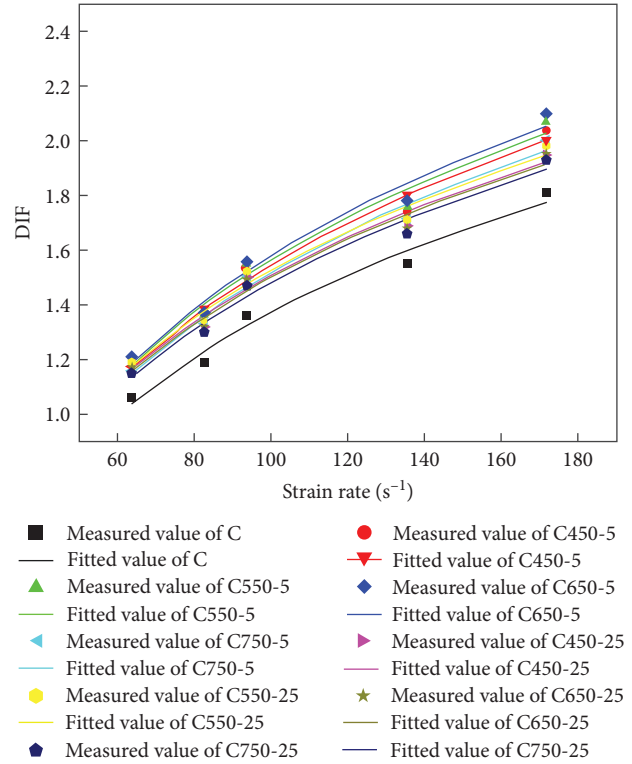


FIGURE 5: Comparison of measured DIF values with fitted values.

$$\left\{ \begin{array}{ll} \text{C:} & \text{DIF} = 1.70959 \lg \dot{\epsilon} - 2.04532 \quad R^2 = 0.9801 \\ \text{C425-5:} & \text{DIF} = 1.93953 \lg \dot{\epsilon} - 2.33392 \quad R^2 = 0.9818 \\ \text{C525-5:} & \text{DIF} = 1.96649 \lg \dot{\epsilon} - 2.36815 \quad R^2 = 0.9790 \\ \text{C625-5:} & \text{DIF} = 2.01447 \lg \dot{\epsilon} - 2.44868 \quad R^2 = 0.9789 \\ \text{C725-5:} & \text{DIF} = 1.89495 \lg \dot{\epsilon} - 2.27224 \quad R^2 = 0.9835 \\ \text{C425-25:} & \text{DIF} = 1.75289 \lg \dot{\epsilon} - 1.99843 \quad R^2 = 0.9808 \\ \text{C525-25:} & \text{DIF} = 1.78941 \lg \dot{\epsilon} - 2.05192 \quad R^2 = 0.9807 \\ \text{C625-25:} & \text{DIF} = 1.76374 \lg \dot{\epsilon} - 2.02627 \quad R^2 = 0.9813 \\ \text{C725-25:} & \text{DIF} = 1.76374 \lg \dot{\epsilon} - 2.04627 \quad R^2 = 0.9813 \end{array} \right. \quad (3)$$

3.4.3. Energy. Energy absorption is an important parameter to measure the impact load resistance of concrete materials [39]. The absorption energy is calculated as shown in Formula (4):

$$W_A(t) = EC \int (\epsilon_I^2(t) - \epsilon_R^2(t) - \epsilon_T^2(t)) dt, \quad (4)$$

where E is the elastic modulus of the bar, C is the impact velocity, and $\epsilon_I(t)$, $\epsilon_R(t)$, and $\epsilon_T(t)$ are incident strain, reflected strain, and transmitted strain, respectively.

The calculated absorbed energy results are presented in Figure 6. It is evident from Figure 6 that as the impact

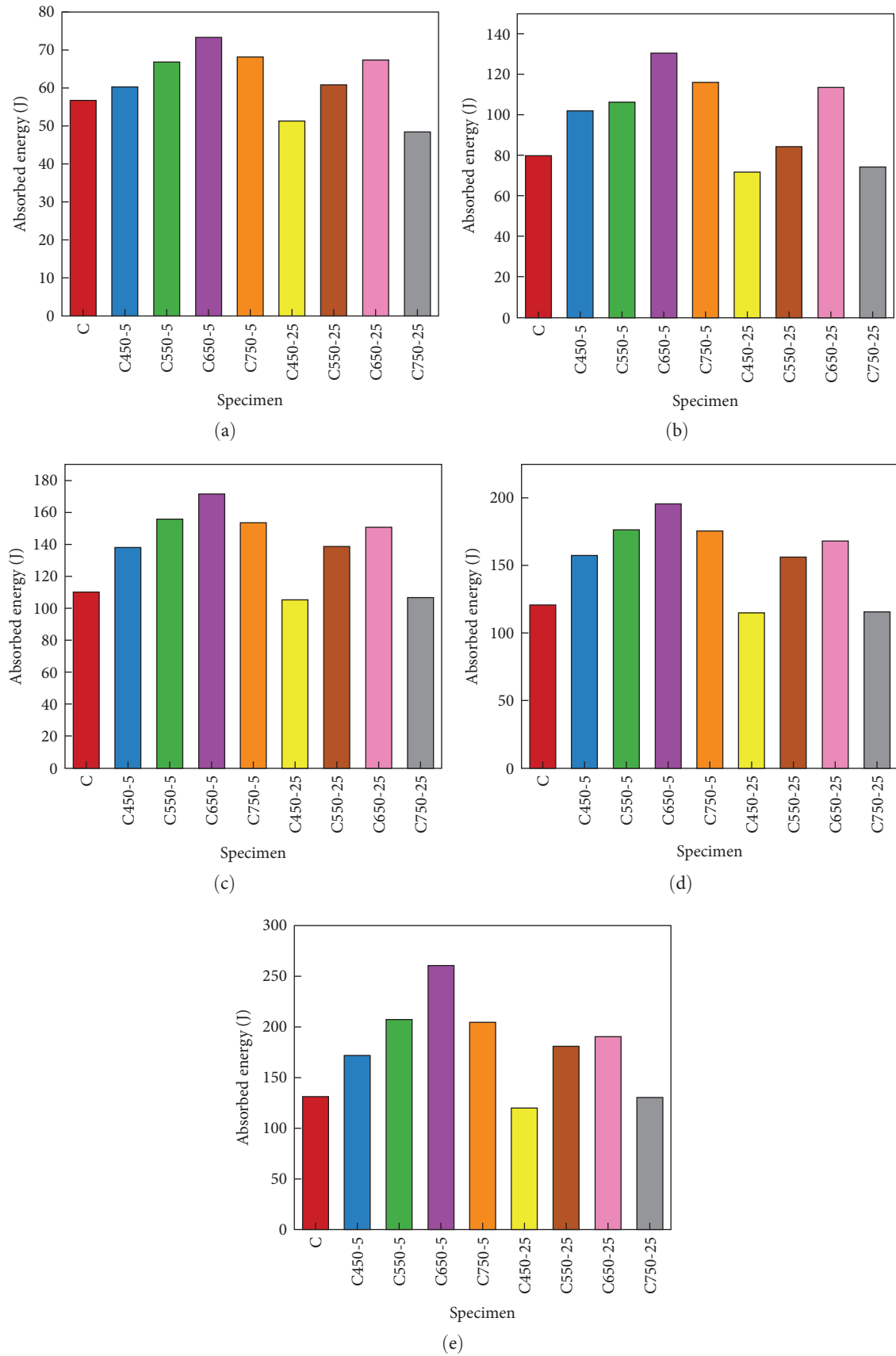


FIGURE 6: Absorbed energy at different impact pressures: (a) 0.25 MPa, (b) 0.35 MPa, (c) 0.45 MPa, (d) 0.55 MPa, and (e) 0.65 MPa.

TABLE 3: Statistical results of input data and output data.

Variable	Min value	Max value	Standard deviation	Variance
Cement (%)	90	100	3.18	10.10
Rice husk ash (%)	0	10	3.18	10.10
Calcination temperature (°C)	0	750	218.47	47,727.27
Grain size (μm)	0	25	10.66	113.64
Impact load (MPa)	0.25	0.65	0.14	0.02
Dynamic compressive strength (MPa)	30.48	76.34	11.11	123.40

pressure increases, the absorbed energy of the nine sample groups rises. For example, at an impact pressure of 0.55 MPa, compared to 0.25 MPa, the absorbed energy of C450-5, C550-5, C650-5, C750-5, C450-25, C550-25, and C650-25 increased by 112.5%, 161%, 164%, 167%, 156.5%, 143.5%, and 156.5%, respectively. This result aligns with previous findings [40, 41]. This is mainly because with the increase of impact pressure, concrete materials need to absorb more energy to resist external loads, so the absorbed energy gradually increases [42]. It should be pointed out that under the same conditions, the absorption energy of C450-25 and C750-25 is lower than that of ordinary concrete. In addition, when the calcination temperature of rice husk ash is 650°C, the absorption energy performance is the best.

3.5. Dynamic Compression Strength Prediction Based on Machine Learning. While experimental methods provide precise measurements of concrete properties, machine learning predictions help identify nonlinear relationships and optimize parameters that are difficult to explore exhaustively through experiments alone. Additionally, machine learning allows for efficient predictions of concrete properties under conditions not explicitly tested in experiments, thereby reducing the need for extensive experimental trials. This complementary approach bridges gaps in experimental insights and provides a more holistic understanding of material behavior.

3.5.1. Establishment of Data Set. To evaluate the effects of calcination temperature and rice husk ash particle size on the dynamic compressive strength of concrete, this study selected cement percentage, rice husk ash percentage, calcination temperature, rice husk ash particle size, and impact pressure as input variables, with dynamic compressive strength as the output variable. In this study, the ratio of the training set to the test set was 8:2. Luo et al. [43] emphasized that data distribution plays a significant role in the performance of machine learning models. The statistical results of input and output variables are provided in Table 3.

The Seaborn Python library was employed to analyze the relationship between input and output variables, and the results are presented in Figure 7. Figure 7 illustrates that the initial effects of input variables on output variables are easily observed. For instance, the mean particle size of rice husk ash shows a negative correlation with dynamic compressive strength. Another example is the positive correlation between impact pressure and dynamic compressive strength. In addition, there is an obvious nonlinear relationship between other variables and dynamic compressive strength. Of course, the relationship between the

final input variable and the output variable needs to be verified by machine learning techniques.

The relationship between parameters and indicators plays a crucial role in the model's prediction accuracy. In this study, the Pearson correlation coefficient was employed to assess the linear relationships between variables, and the calculation method is provided in Equation (5). The PCC values for the variables are displayed in Figure 8. As shown in Figure 8, most input variables have PCC values below 0.25, indicating that no significant multicollinearity exists between the variables, thus preventing any negative impact on the prediction results:

$$PCC = \frac{\sum_{i=1}^N (y_i - \bar{y})(y'_i - \bar{y}')}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (y'_i - \bar{y}')^2}}. \quad (5)$$

3.5.2. Introduction of eXtreme Gradient Boosting (XGB) Model. The XGB model is an ensemble learning method commonly used for classification and regression tasks. The XGB model has a strong capability to process features and capture nonlinear relationships, making it highly effective in predicting concrete strength. Here is a brief explanation of the XGB model:

1. The XGB model is based on a gradient lifting algorithm that builds a strong learner by combining multiple weak learners (usually decision trees). Each new tree is built to correct the prediction errors of all previous trees, thereby continuously improving the prediction accuracy of the model.
2. The training process involves the following steps: First, the model is initialized with a simple baseline model (such as a constant model). Then, iterative training is conducted to calculate the current model's prediction error, and a new decision tree is constructed based on this error. The goal is to minimize the current model's error, after which the new tree is integrated into the existing model, and finally, the model output, through the combination of multiple trees, the final output of the prediction result.

3.5.3. Hyperparameter Optimization. Hyperparameter optimization is a crucial step in machine learning. Common methods for hyperparameter tuning include grid search, random search, and Bayesian optimization [44]. The basic principle of grid search is to discretize the hyperparameters to be

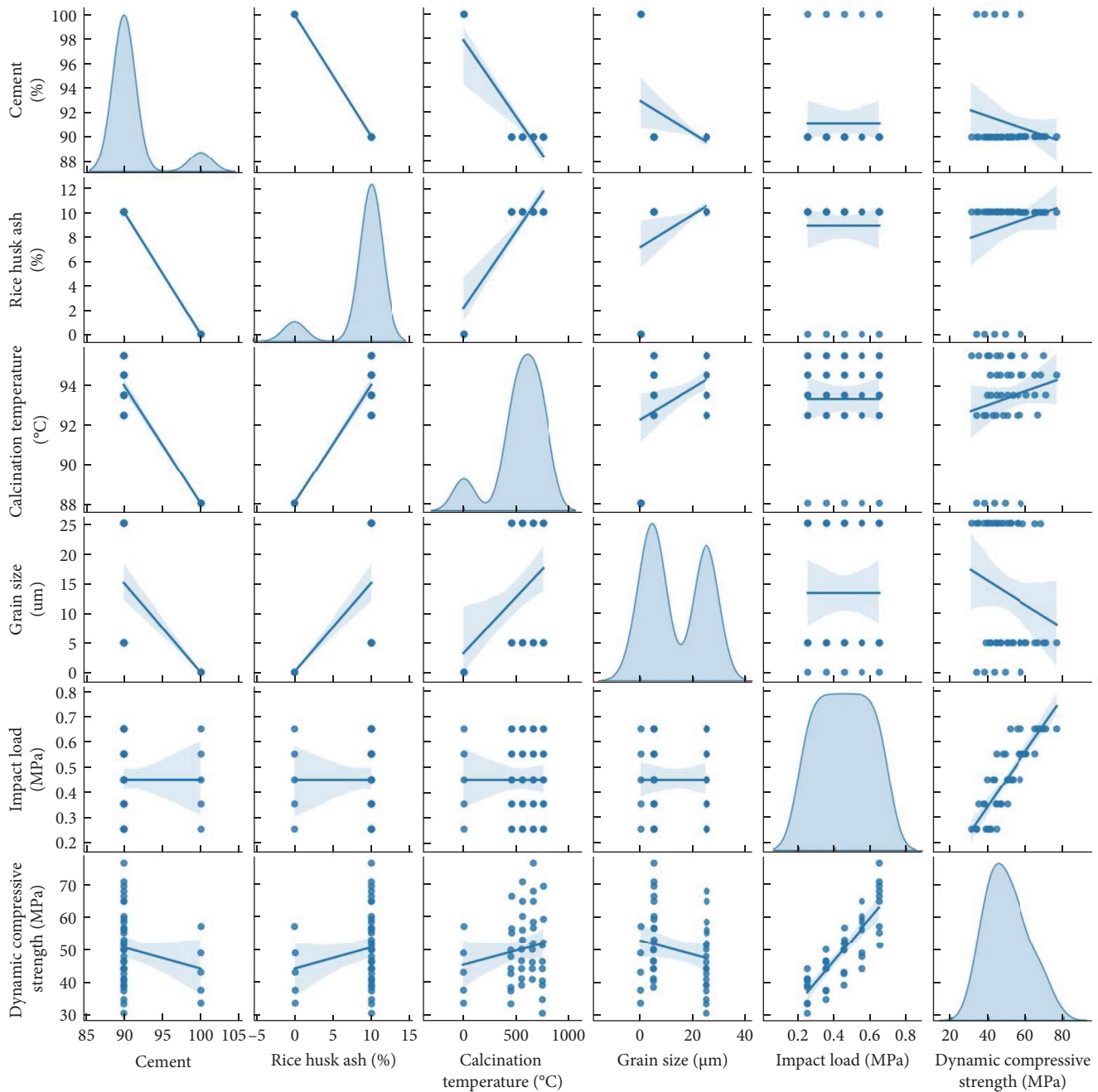


FIGURE 7: Distribution of variables.

adjusted according to a certain range and step size to form a grid and then train and evaluate each combination in the grid to find the optimal combination of hyperparameters [45]. Compared with the other three hyperparameter optimization methods, grid search is relatively simple to implement and understand and does not depend on any assumptions about the hyperparameter space, so there is no need for any prior knowledge about the distribution of the parameter space [46]. In addition, since the training and evaluation of each parameter combination are independent of each other, grid search can be easily parallelized, thus speeding up the search

speed [46]. In this study, grid search is used to optimize the hyperparameters, and the results are shown in Table 4.

3.5.4. Prediction Results. Figure 9 presents the error data from dynamic compression experiments and XGB model predictions. As shown in Figure 9, the XGB model's predictions closely align with the experimental results, with an overall error not exceeding 10 MPa. This demonstrates that the XGB model is effective in predicting dynamic compressive strength in this study and highlights its broad forecasting capabilities.

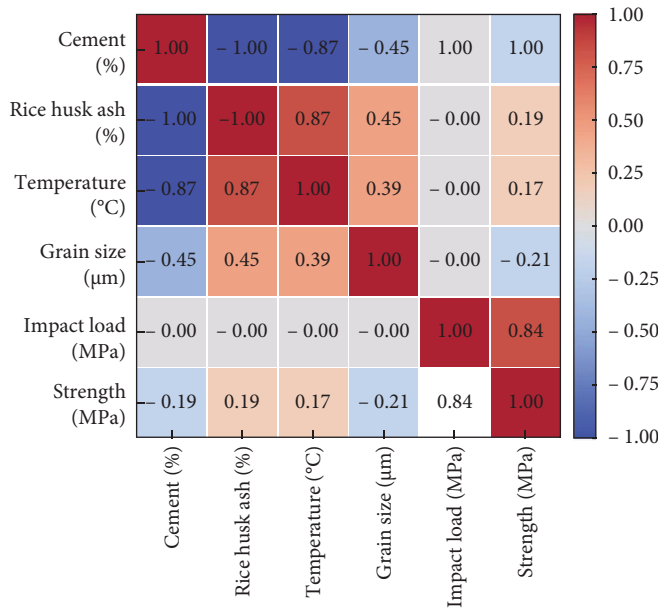


FIGURE 8: Correlation coefficient of variables.

TABLE 4: Results of hyperparameter optimization of XGB model.

Parameter	Values range	Value
n_estimators	10–1000	1000
max_depth	1–100	1
Gamma	0.01–1	0.0372759372031494

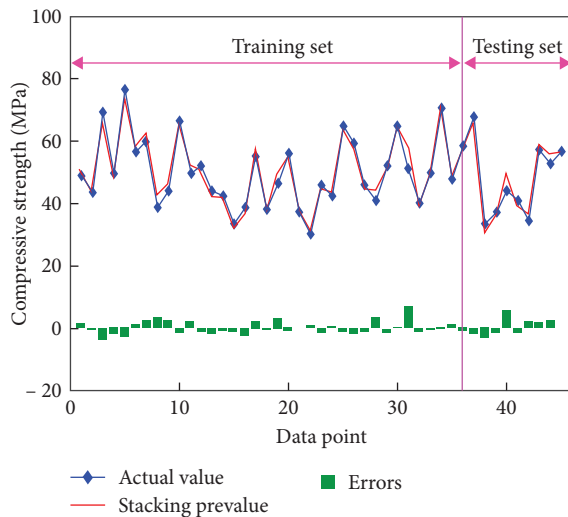


FIGURE 9: Comparison of error information between experimental data and XGB model prediction data.

The prediction performance metrics of the XGB model were calculated, including coefficient of determination (R^2), mean absolute error (MAE), mean square error (MSE), root mean square error (RMSE), explained variance (EV), and maximum variance (ME). The results are presented in Figure 10. As shown in Figure 10, the R^2 values for the

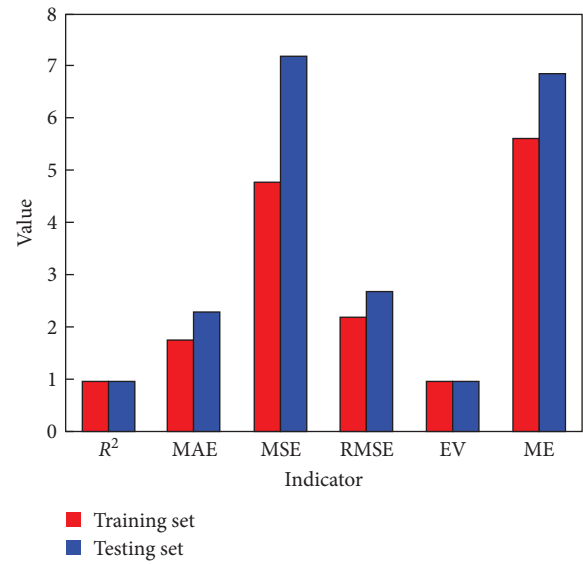


FIGURE 10: Performance indicators.

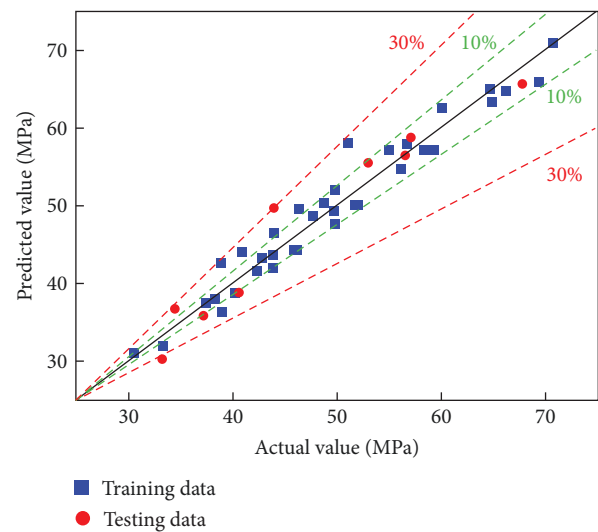


FIGURE 11: Scatter plot of actual dynamic compression strength and predicted value.

training and test data sets are 0.9592 and 0.9435, respectively. This demonstrates the high prediction accuracy of the XGB model for both data sets. Additionally, the training set's MAE, MSE, RMSE, and ME were 1.7539, 4.7667, 2.1833, and 5.6018, respectively, while the test set's values were 2.2863, 7.1912, 2.6816, and 6.8712. Despite the larger deviation between predicted and measured values in the test data compared to the training data, the results remain highly reliable. Furthermore, the EV values for both the training and test data sets were 0.9592 and 0.9449, respectively, both above 0.9, indicating the XGB model's strong performance in predicting dynamic compressive strength.

Figure 11 shows the scatter plot of experimental data and predicted data. As can be seen from Figure 11, although a small number of training and test data sets have data point

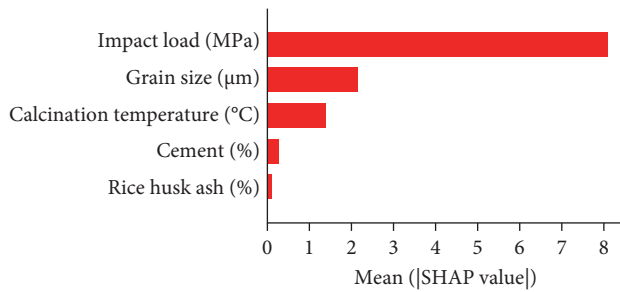


FIGURE 12: Average SHAP values for input variables.

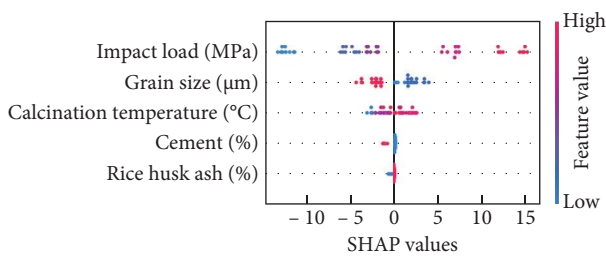


FIGURE 13: SHAP influence value.

errors greater than 10%, most data points are within 10%. It shows that the predicted value obtained in this study has good accuracy and the data point dispersion is small.

3.6. Influence Analysis of Dynamic Compression Strength Based on SHapley Additive exPlanations (SHAP)

3.6.1. Global Analysis. To examine the correlation of input variables, this study employed SHAP [47, 48] to analyze the impact of input variables on dynamic compressive strength. Figure 12 presents the average SHAP values for each variable. As shown in Figure 12, impact pressure has the most significant effect on dynamic compressive strength, with an average SHAP value exceeding 8.0. It is also noteworthy that the influence of rice husk ash particle size on dynamic compressive strength is greater than that of calcination temperature [49].

In this study, SHAP heat map was used to explain the influence of input variables on the dynamic compression strength of each sample, and the results were shown in Figure 13. In Figure 13, red and blue indicate positive and negative effects, respectively. It is observed that the impact pressure has a positive effect on the dynamic compressive strength, while the increase of the average particle size of rice husk ash has a negative effect on the dynamic compressive strength. In addition, the influence of calcination temperature of rice husk ash on dynamic compressive strength is nonlinear; that is, there is an optimal calcination temperature [50, 51].

3.6.2. Impact Analysis of a Single Input Variable. To better understand the impact of a single input variable on dynamic compressive strength, Figure 14 illustrates the SHAP influence diagram for one input variable. It can be clearly seen from Figure 14 the influence trend of a single input variable on the dynamic compression strength. For example, using rice husk ash to replace part of cement in concrete and reducing

the average particle size of rice husk ash are beneficial to dynamic compressive strength. For example, with the increase of the calcination temperature of rice husk ash, the dynamic compressive strength increases first and then decreases.

4. Conclusion

The impact of rice husk ash particle size and calcination temperature on concrete properties such as slump, water absorption, compressive strength, and dynamic compression was investigated by substituting 10% of the cement with rice husk ash. Furthermore, the dynamic compressive strength of rice husk ash concrete was forecasted using the XGB machine learning model, with SHAP sensitivity analysis conducted. The key findings are summarized as follows:

1. The addition of rice husk ash decreases concrete slump. Concrete performance results show that as the calcination temperature of rice husk ash increases, the slump of the concrete decreases. Additionally, a clear reduction in slump was observed with smaller rice husk ash particle sizes.
2. The inclusion of rice husk ash raises the water absorption rate of concrete. When the particle size of rice husk ash remains constant, the water absorption initially decreases and then increases with rising calcination temperatures. Smaller particle sizes of rice husk ash result in higher water absorption.
3. When the particle size of rice husk ash is consistent, the compressive strength, dynamic compressive strength, and absorption energy reach their peak at a calcination temperature of 650°C. As the calcination temperature increases, the DIF value of rice husk ash first rises and then declines. Additionally, the DIF value decreases as particle size increases.
4. A comparison between predicted results from the XGB machine learning model and experimental data indicates good accuracy and fit. The analysis of input variables reveals that impact pressure has the greatest influence on dynamic compressive strength, with particle size having more effect than calcination temperature. Moreover, when the calcination temperature of rice husk ash exceeds 650°C, it negatively affects strength.
5. These findings suggest that rice husk ash concrete has significant potential for practical applications in green construction and engineering. Its ability to reduce energy consumption, lower greenhouse gas emissions, and promote agricultural waste recycling positions it as a promising material for sustainable infrastructure development.

While the environmental benefits of rice husk ash concrete are evident, its relevance to broader construction practices and industry applications warrants further emphasis. The use of rice husk ash as a partial cement replacement aligns with global efforts to reduce the carbon footprint of construction activities. For instance, the reduced cement

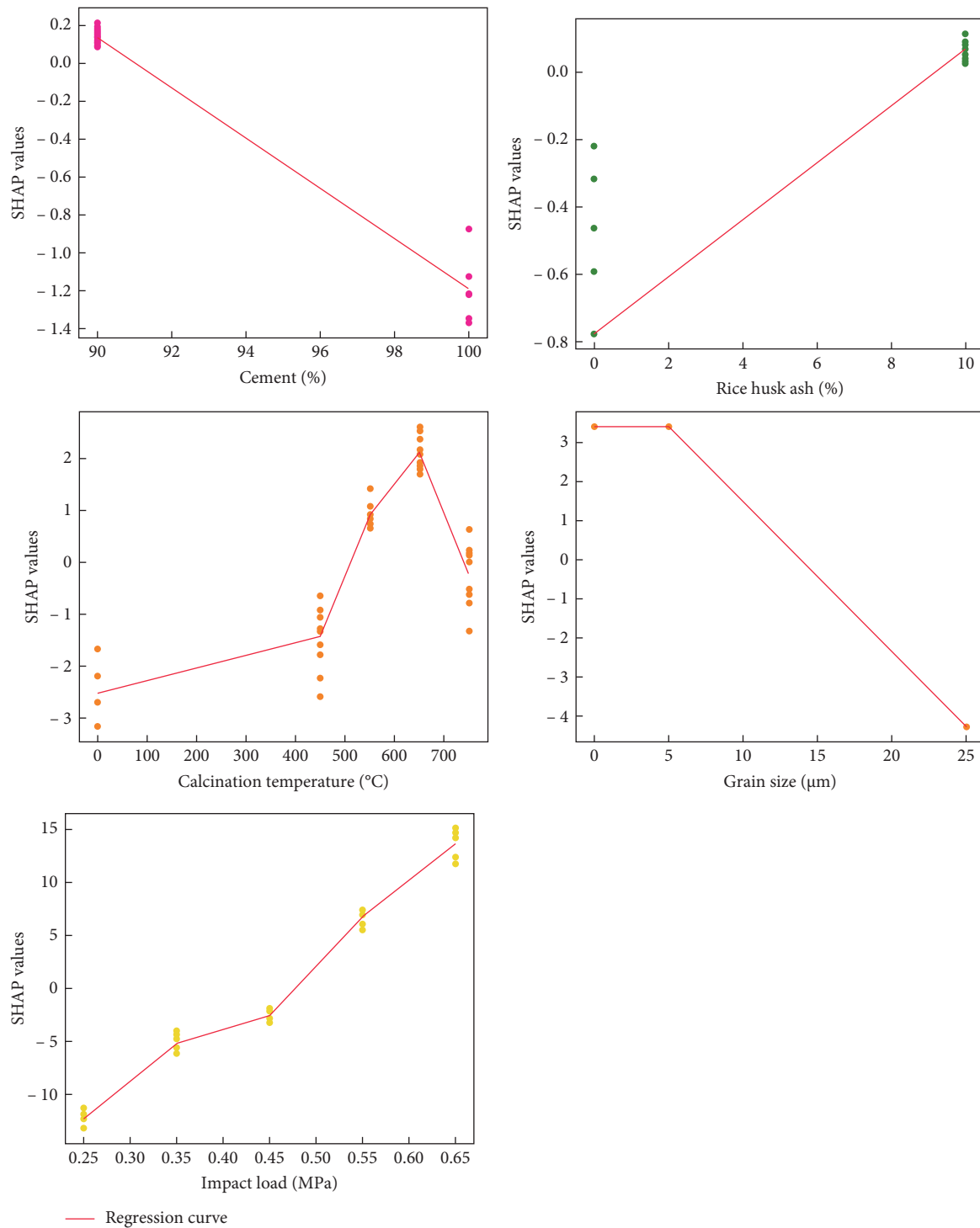


FIGURE 14: Sensitivity of a single input variable to dynamic compression strength.

consumption directly contributes to lower greenhouse gas emissions, while the recycling of agricultural waste addresses critical issues of resource depletion and waste management. Moreover, rice husk ash concrete holds promise for applications in green building certifications, sustainable infrastructure projects, and low-carbon urban development. Its cost-effectiveness and performance make it a suitable material for large-scale projects, such as highway pavements,

prefabricated building components, and water-resistant structures in flood-prone regions.

This study focused on calcination temperatures ranging from 450°C to 750°C and particle sizes of 5 and 25 μm. However, to better understand the broader effects of rice husk ash on concrete performance, future studies could explore calcination temperatures outside this range, such as lower temperatures (300°C–400°C) to investigate the partial

carbonization phase or higher temperatures (800°C–900°C) to study the effects of silica crystallization. Similarly, expanding the particle size range to include finer particles (1–5 µm) could provide insights into nanoscale effects, while larger particle sizes (25–50 µm) may reveal trade-offs between workability and strength in practical applications. These extended parameters could offer a more comprehensive understanding of the mechanisms influencing rice husk ash concrete properties, as well as its potential for tailored applications in construction. In addition to the mechanical and workability properties studied in this research, future work could explore other critical performance metrics of rice husk ash concrete to broaden its application potential. For example, investigating its durability under varying environmental conditions, such as freeze–thaw cycles, sulfate attack, and carbonation, would provide insights into its long-term performance and resilience in diverse climates.

Data Availability Statement

The data used to support the findings of this study are included within the article.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

This research was supported by the Key Scientific Research Projects of Universities in Henan Province under grant number 24B560020.

Acknowledgments

This research was supported by the Key Scientific Research Projects of Universities in Henan Province under grant number 24B560020.

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